

Oppgaver for group sessions uke 13 (23.03-27.03)

MNF130

Spring 2026

Oppgavene er hentet fra læreboken K. Rosen: Discrete Mathematics and its Applications, kapittel 4.3, 5.1.

1. Determine whether each of these integers is prime: 21, 29, 71, 111.
2. Find the prime factorization of these integers: 88, 126, 729, 1001.
3. Show that $a^m + 1$ is composite if a and m are integers greater than 1 and m is odd. [*Hint*: Show that $x + 1$ is a factor of the polynomial $x^m + 1$ if m is odd.]
4. The value of the **Euler ϕ (phi) function** at the positive integer n is defined to be the number of positive integers less than or equal to n that are relatively prime to n . For instance, $\phi(6) = 2$ because, of the positive integers less than or equal to 6, only 1 and 5 are relatively prime to 6.
 - (a) Find these values of the Euler ϕ function: $\phi(4)$, $\phi(10)$, $\phi(13)$.
 - (b) Show that n is prime if and only if $\phi(n) = n - 1$.
 - (c) What is the value of $\phi(p^k)$ when p is prime and k is a positive integer?
5. Use the Euclidean Algorithm to find:
 - (a) $\gcd(12, 18)$.
 - (b) $\gcd(111, 201)$.
 - (c) $\gcd(1001, 1331)$.
6. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for the positive integer n .
 - (a) What is the statement $P(1)$?
 - (b) Show that $P(1)$ is true, completing the basis step of a proof that $P(n)$ is true for all positive integers n .
 - (c) What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?

- (d) What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
 - (e) Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you used the inductive hypothesis.
 - (f) Explain why these steps show that this formula is true whenever n is a positive integer.
7. Prove that $2 + 2(-7) + 2(-7)^2 + \cdots + 2(-7)^n = \frac{1 - (-7)^{n+1}}{4}$ whenever n is a non-negative integer.
8. Prove that

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

whenever n is a non-negative integer.